

FYP DEFENSE PREPARATION

Questions & Answers Guide

AI-Based System for Detection of Manipulated Media Content

FYP ID: 26-FYP-229

Final Year Project — BS Software Engineering (2022–2026)

Prepared for: Supervisor, Co-Supervisor, Internal & External Committee

How to use this document

This guide compiles the most likely questions a Supervisor, Co-Supervisor, Internal Committee, and External Committee member would ask during an FYP evaluation or defense, organized by topic rather than by panel role, since most questions can come from any examiner regardless of title. Each question includes a complete answer, the justification or reasoning behind key decisions, and where relevant, the research or references that informed the choice. Read through each section before your defense, and practice answering in your own words rather than memorizing verbatim — panels notice rehearsed answers and often follow up with a variant of the same question.

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Section 1: Project Motivation, Problem Statement & Scope

These questions establish whether you understand WHY this project exists and what real-world problem it solves. Examiners often start here to gauge whether you can articulate motivation beyond "it was a good FYP topic."

Q1. Why did you choose deepfake detection as your FYP topic? *[Motivation]*

A: Deepfake technology has grown rapidly since 2017, and AI-generated manipulated media now poses a real threat to journalism, elections, personal reputations, and general digital trust. We noticed that most existing detection tools fall into two extremes: either they are academic research prototypes that are accurate but require deep technical expertise and expensive GPU hardware to run, or they are commercial tools that are accurate but operate as black boxes and charge high subscription fees. We wanted to build something that sits in the middle — accurate, but also explainable, accessible on standard hardware, and usable by someone with no AI background.

Q2. What is the core problem your system solves? *[Problem Statement]*

A: The core problem is that ordinary people — journalists, educators, social media users — currently have no easy way to verify whether a video has been digitally manipulated. Existing solutions are either inaccessible (require technical setup, GPUs, or coding knowledge), opaque (give a yes/no answer without explaining why), or costly. Our system solves this by providing a free, web-based platform where any user can upload a video and receive not just a verdict, but a confidence score, a visual heatmap showing exactly which regions of the face look suspicious, and a plain-language explanation.

Q3. Who is the target user of this system? *[Scope]*

A: We designed the system for four main user groups, each with a slightly different need:

- Journalists and fact-checkers who need to quickly verify video authenticity before publishing.
- Educators who want to teach students about media literacy and demonstrate how deepfakes are detected.
- Everyday social media users who encounter a suspicious video and want a quick, free way to check it.
- Researchers and students who want an open-source reference implementation to build on.

Q4. What is out of scope for this project? *[Scope]*

A: Several things were deliberately excluded to keep the project achievable within an FYP timeline:

- Real-time video stream analysis (e.g., live video calls) — we only support uploaded video files.
- Audio deepfake / voice cloning detection — our model is purely visual/frame-based.
- Mobile native applications — we built a responsive web app, not iOS/Android apps.
- Training our own dataset from scratch — we used existing benchmark datasets rather than collecting new data.

These are explicitly listed as Future Work in Chapter 10.

Q5. How is this different from just using an existing tool like Microsoft Video Authenticator or Deepware?*[Comparative Analysis]*

A: Those tools either restrict access (Microsoft's tool was limited to select partners and not publicly available), or operate as a closed commercial product without explainability (Deepware gives a score but no visual or textual reasoning). Our comparative analysis in Chapter 10 shows that we are the only system in our comparison set that combines multimodal analysis, explainable AI output (heatmap + text), CPU-only execution, a complete web application, and free/open-source availability all in one platform.

Section 2: Project Objectives

Examiners frequently ask you to justify or expand on specific objectives, or to point out where in the system a particular objective is fulfilled. Be ready to map each objective to a concrete feature.

Q6. You have 12 objectives — isn't that too many for one FYP? *[Objectives]*

A: We organized them into 5 primary objectives (the core detection system: AI detection, user interface, explainability, complete system, CPU optimization) and 7 secondary objectives (supporting features: admin panel, format support, documentation, accuracy/transparency balance, scalability, AI chat assistants, and the subscription model). The primary objectives represent the minimum viable product; the secondary objectives reflect features we added as the project matured, particularly the AI assistants and subscription system, which were extensions added after the initial proposal was approved.

Q7. Which objective do you consider the most important, and why? *[Objectives]*

A: Objective 3 — implementing explainable AI with confidence scores, heatmaps, and explanations — is the most important, because it is what differentiates our system from every black-box alternative on the market. A detection system that says 'fake' or 'real' without justification is not trustworthy in high-stakes contexts like journalism. Our entire design, from the Grad-CAM heatmap generation to the Analysis Assistant chatbot, was built to support this single objective.

Q8. How did your objectives change from your initial proposal to the final version? *[Objectives Evolution]*

A: Our initial proposal focused on the five primary objectives: detection accuracy, usability, explainability, a complete system, and CPU optimization. As development progressed, we identified two valuable extensions that supported these same goals: AI-powered conversational assistants (to make explainability more interactive — Objective 11) and a tiered subscription model (to make the system financially sustainable for continued hosting and scaling — Objective 12). Both were added as secondary objectives with supervisor approval, not as scope creep, since they reinforce rather than replace the original goals.

Q9. How will you measure whether you achieved your objectives? *[Validation]*

A: Each objective maps to a specific, testable outcome documented in Chapters 7 and 10:

- Detection accuracy — measured via model evaluation on test datasets and reflected in confidence scores.
- Usability — validated through the use case and test case coverage in Chapter 3 and Chapter 7 (17 test cases, all passing).
- Explainability — demonstrated through the heatmap and textual explanation features, shown in the Application Prototype chapter.
- CPU accessibility — demonstrated by deploying ConvNeXt-Tiny for inference without requiring GPU acceleration.
- Scalability — addressed through the three-tier architecture design which allows independent scaling of each layer.

Section 3: AI Model, Detection Approach & Explainability

This is typically the most technically scrutinized section, especially by the external committee. Expect deep follow-up questions on model choice, accuracy, and limitations.

Q10. Why did you choose ConvNeXt-Tiny specifically, instead of ResNet, EfficientNet, or a Vision Transformer? [Model Choice]

A: ConvNeXt-Tiny was chosen because it incorporates modern architectural improvements inspired by Vision Transformers — such as larger convolutional kernel sizes and improved normalization strategies — while remaining a pure CNN, which keeps inference more efficient on CPU than a full transformer architecture would be. We initially experimented with EfficientNet-B4 using a spatial attention mechanism for binary (real/fake) classification, but found that moving to ConvNeXt-Tiny with 6-class classification on FaceForensics++ gave us more granular and informative detection output — identifying the specific manipulation technique rather than a simple real/fake label — which better supports our explainable AI objective.

Q11. Why didn't you use a Vision Transformer (ViT), and why move away from your earlier EfficientNet-B4 model? [Model Choice / Limitations]

A: Vision Transformers do show strong performance on large-scale image classification benchmarks, but they typically require significantly more training data and computational resources to reach that performance, and inference is generally slower on CPU compared to CNNs of similar accuracy. Since one of our core objectives was CPU-accessible deployment without expensive hardware, a CNN-based architecture remained the more pragmatic choice. As for EfficientNet-B4: our earlier iteration used it with a spatial attention mechanism for binary real/fake classification, and it performed adequately, but a binary label has limited explanatory value for the user — it doesn't indicate what kind of manipulation was detected. ConvNeXt-Tiny trained for 6-class classification let us report the specific manipulation technique (Deepfakes, Face2Face, FaceShifter, FaceSwap, or NeuralTextures) alongside the genuine/original class, which directly strengthened our explainability objective. We note transformer-based architectures as a direction for Future Work in Chapter 10, where additional GPU resources could be explored.

Q12. How does your detection pipeline actually work, step by step? [Technical Implementation]

A: The pipeline follows five sequential stages, all implemented in the backend's video processing service:

- 1. Frame extraction — frames are extracted from the uploaded video at a fixed sampling rate (1 frame per second) rather than processing every frame, to balance accuracy with processing time.
- 2. Face detection — each extracted frame is scanned for faces using a face detection algorithm.
- 3. Feature extraction — visual features are extracted from each detected face using the ConvNeXt-Tiny convolutional layers.
- 4. Inference — the model runs a forward pass on each face to produce a per-frame prediction across six classes: five manipulation types (Deepfakes, Face2Face, FaceShifter, FaceSwap, NeuralTextures) and one genuine/original class.

- 5. Aggregation — predictions across all frames are aggregated into a single video-level confidence score, and Grad-CAM is applied to generate the heatmap visualization for the most suspicious frame(s).

Q13. What is Grad-CAM, and why did you use it instead of another explainability technique? *[Explainable AI]*

A: Grad-CAM (Gradient-weighted Class Activation Mapping) is a technique that uses the gradients flowing into the final convolutional layer of a CNN to produce a heatmap highlighting which regions of the input image most influenced the model's prediction. We chose it over alternatives like LIME or SHAP because Grad-CAM is specifically designed for CNN architectures and integrates naturally with PyTorch without requiring additional perturbation-based sampling, which would significantly slow down inference. It also produces visually intuitive heatmaps that non-technical users can immediately understand — directly supporting our usability and explainability objectives.

Q14. What datasets did you use to train and evaluate your model? *[Training Data]*

A: We used FaceForensics++, a publicly available benchmark dataset that is a standard reference in deepfake detection research (as discussed in our Literature Review, Chapter 2). FaceForensics++ conveniently provides the five manipulation-type labels used in our 6-class classification scheme — Deepfakes, Face2Face, FaceShifter, FaceSwap, and NeuralTextures — alongside genuine, unmanipulated footage, allowing the model to learn to distinguish between specific manipulation techniques rather than just a generic real/fake distinction. An earlier iteration of the model was also evaluated using the DeepFake Detection Challenge dataset for binary classification before we moved to the 6-class FaceForensics++-based approach.

Q15. What is your model's accuracy, and how confident are you in that number? *[Model Performance — ACTION REQUIRED]*

A: [Note to team: insert your actual measured accuracy/precision/recall/F1 figures for the ConvNeXt-Tiny 6-class model before the defense — this answer must reflect your real test results, not a placeholder. Also be ready to discuss per-class performance, since panels often ask this specifically for multi-class models: is accuracy roughly even across all six classes (the five manipulation types plus genuine), or does the model perform noticeably better on some manipulation techniques than others? Be ready to explain what dataset split was used for evaluation (train/validation/test) and whether the test set included manipulation techniques not seen during training.]

Q16. What are the limitations of your detection model? *[Limitations]*

A: We are transparent about several limitations:

- The model analyzes visual artifacts only — it does not detect audio-based manipulation or voice cloning.
- Performance may degrade on deepfake generation techniques not represented in the training data, since deepfake generators continue to evolve.
- Frame sampling at 1fps means very brief manipulated segments between sampled frames could potentially be missed.
- CPU-only inference, while accessible, is slower than GPU-accelerated inference, which affects processing time for longer videos.

Q17. How do you handle videos with multiple faces or no detectable face? *[Edge Cases]*

A: For videos with multiple faces, the system runs face detection independently per frame and analyzes each detected face; the aggregation step considers predictions across all detected faces when producing the final score. If no face is detected in a frame, that frame is excluded from analysis rather than producing a false prediction. If no faces are detected across the entire video, the system returns an appropriate error/notification to the user rather than a misleading result, since the model's analysis specifically depends on facial regions for both prediction and heatmap generation.

Section 4: Development Methodology

Q18. Why did you choose the Waterfall Model instead of Agile or Scrum? *[Methodology]*

A: The Waterfall Model was chosen because the core requirements of the system — dataset selection, model architecture, backend API, and frontend interface — could be reasonably well-defined at the outset of the project, which is the condition under which Waterfall performs best. It also matches the academic structure of an FYP, where comprehensive documentation at each phase (requirements, design, testing) is itself a deliverable, not just a means to an end. Additionally, the phase-gated structure reduced the risk of rework, since requirements and architecture were finalized before implementation began.

Q19. Isn't Waterfall outdated for software projects? Why not Agile, which is industry standard? *[Methodology Justification]*

A: Agile is indeed more common in industry, particularly for products with evolving or unclear requirements. However, our project's requirements were largely fixed early on — we knew we needed a detection model, an API, and a frontend, with clear specifications for each. Agile's strength (rapid iteration in response to changing requirements) was less critical for us than Waterfall's strength (structured documentation matching academic deliverable expectations and clear phase-to-phase traceability for evaluation). It is also worth noting that even with Waterfall as the overarching model, we still used iterative development practices within each phase, such as testing and refining the model multiple times during the Implementation phase.

Q20. How did you manage the project timeline across team members? *[Project Management]*

A: The Schedule of Activities (Chapter 1) breaks the 20-week timeline into five phases — Planning & Research (4 weeks), AI Model Development (6 weeks), Backend Development (3 weeks), Frontend Development (3 weeks), and Testing & Deployment (4 weeks) — each with defined milestones (Design Complete, AI Model Ready, Backend Complete, Frontend Complete, Project Deployed). Team members were assigned primary responsibility for different phases based on their strengths, while still collaborating across phases, particularly during integration testing where backend, frontend, and AI components had to work together.

Section 5: System Architecture & Technology Stack

Q21. Why did you choose a three-tier architecture? *[Architecture]*

A: A three-tier architecture (Presentation, Application, Data) provides clear separation of concerns, which directly supports our objectives of testability, scalability, and maintainability. Each tier can be developed, tested, and scaled independently — for example, if the AI processing load grows, the Application tier can be scaled without needing to change the frontend or database layer. It also improves security, since direct database access is restricted to the Application tier rather than exposed to the Presentation layer.

Q22. Why FastAPI over Django or Flask for the backend? *[Technology Choice]*

A: FastAPI was chosen for three main reasons: it supports asynchronous request handling natively, which is important given that video analysis is a long-running, resource-intensive operation; it auto-generates interactive API documentation (Swagger UI), which sped up our own development and testing; and it offers performance closer to Node.js-based frameworks than traditional synchronous Python frameworks like Flask or Django, due to its ASGI foundation built on Starlette and Pydantic for data validation.

Q23. Why Next.js instead of plain React or another frontend framework? *[Technology Choice]*

A: Next.js was chosen for its built-in server-side rendering (SSR) and static generation capabilities, which improve initial page load performance and SEO compared to a plain client-side React app — relevant since our landing page needs to be discoverable and fast-loading for new visitors. It also provides a built-in routing system and API route conventions that simplified our frontend architecture, while still being fully React-based underneath, so we retained React's component model and ecosystem.

Q24. Why PostgreSQL instead of MongoDB or another NoSQL database? *[Technology Choice]*

A: Our data is inherently relational — users have videos, videos have frames, and these relationships benefit from foreign key constraints and referential integrity, which PostgreSQL enforces natively. We also needed ACID-compliant transactions, particularly for subscription and payment data where partial or inconsistent writes could cause real financial/account problems. PostgreSQL additionally supports JSON column types, which we used for flexible fields like `analysis_details` and `prediction_details`, giving us relational structure where it matters and flexibility where it helps, without needing a separate NoSQL system.

Q25. Walk us through your database schema and why you structured it this way. *[Database Design]*

A: Our schema has three core tables connected by foreign keys:

- **users** — stores account information (email, username, hashed password, role) and, after the subscription feature was added, subscription-related fields (`subscription_plan`, `subscription_status`, `stripe_customer_id`, etc.).
- **videos** — stores one record per uploaded video, linked to the uploading user via `user_id`, including file metadata, detection status, and the aggregated confidence score and prediction details.
- **frames** — stores one record per analyzed frame within a video, linked via `video_id`, including per-frame fake/suspicious flags, confidence score, and the base64-encoded frame and heatmap images.

This structure lets us store both the video-level summary (for quick dashboard display) and the frame-level detail (for the Analysis Assistant chatbot and detailed result inspection) without duplicating data unnecessarily.

Q26. Why store images as base64 in the database instead of in file storage with just a URL reference?*[Database Design — Be Honest About Trade-offs]*

A: [Note to team: be ready to justify this honestly. A common and valid answer is that base64-in-DB simplified development for the FYP timeline, since it avoided setting up and managing a separate object storage service. If asked about scalability trade-offs, acknowledge that for a production system at larger scale, storing images in dedicated file/object storage (e.g., S3-compatible storage) with the database holding only a reference URL would reduce database size and improve query performance — and mention this as a potential future optimization rather than defending base64 storage as the ideal long-term solution.]

Q27. How does your system ensure data security, particularly for user passwords and payment information? *[Security]*

A: Several measures are in place across different layers:

- Passwords are never stored in plaintext — they are hashed before storage (hashed_password field), so even in the event of a database breach, raw passwords are not exposed.
- Authentication uses JWT (JSON Web Tokens), which are signed and time-limited, reducing the risk of session hijacking compared to long-lived session cookies.
- Payment information is never handled directly by our system — Stripe's hosted Checkout and billing portal handle all sensitive card data, meaning we never store or transmit raw payment card details, which significantly reduces our PCI compliance burden.
- Role-based access control restricts admin-only endpoints (user management, analytics) to users with the admin role.

Section 6: AI Chat Assistants (FAQ & Analysis)

Since this is a newer addition to the project, expect committee members to probe whether it was necessary, how it works technically, and whether it introduces new risks (e.g., hallucination, cost, data privacy).

Q28. Why did you add AI chatbots? Wasn't the heatmap and confidence score already explainable enough? *[Justification]*

A: The heatmap and confidence score are effective for visual explanation, but they are static — a user still has to interpret what the heatmap means on their own. The Analysis Assistant extends explainability by letting users ask natural-language questions like 'why was this flagged as fake?' and receive a conversational answer grounded in the actual frame statistics for their specific video. This makes the explanation interactive rather than passive, which we felt better serves non-technical users who may not know how to interpret a heatmap visually.

Q29. What is the difference between your FAQ Assistant and your Analysis Assistant? *[Architecture]*

A: The FAQ Assistant is available on the public landing page, before login, and answers general questions about the platform (supported formats, how detection works, account requirements) using a static knowledge-base prompt — it does not access the database at all. The Analysis Assistant is only available to authenticated users on their dashboard, and is specific to their own video results — it retrieves real data from the videos and frames tables for the video the user asks about, aggregates statistics, and uses that as grounding context for the AI's response. The FAQ Assistant answers questions about the platform; the Analysis Assistant answers questions about the user's own data.

Q30. What is RAG, and why does your Analysis Assistant use it? *[Technical Approach]*

A: RAG stands for Retrieval-Augmented Generation. Instead of asking a language model to answer purely from its own trained knowledge (which it has no way of knowing about a specific user's video), we first retrieve relevant data from our own database — frame counts, confidence scores, flagged regions — and then pass that retrieved data to the language model as context alongside the user's question. This ensures the AI's response is grounded in real, verifiable data rather than the model guessing or hallucinating an answer, which is critical since users are asking about factual analysis results.

Q31. Why did you choose Groq as your LLM provider instead of OpenAI or Anthropic? *[Technology Choice]*

A: Groq was chosen primarily for two reasons that align with our broader project objectives: it offers a free tier suitable for an FYP's usage volume, and its custom LPU (Language Processing Unit) inference hardware provides very low-latency responses, which keeps the chat experience responsive. This also reinforces our project's recurring theme of accessibility without expensive infrastructure — just as we chose ConvNeXt-Tiny for CPU-accessible detection, we chose Groq for cost-accessible conversational AI.

Q32. What happens if a user asks about a video that doesn't exist, or types the filename wrong? *[Edge Case Handling]*

A: The Analysis Assistant first attempts to extract a filename from the user's message and match it against the videos table for that specific user only (never another user's videos). If no match is found, rather than guessing or hallucinating a response, the system is prompted to either ask the user to provide the exact

filename, or — if the user's intent seems to be browsing rather than asking about a specific video — offer to list all of their analyzed videos so they can identify the correct filename to ask about.

Q33. Do you store chat conversation history? What are the privacy implications? *[Privacy]*

A: No, by design. Conversation history exists only in the frontend's in-memory component state for the duration of the browser session, and is cleared immediately on page refresh or navigation away. We chose this deliberately to keep the implementation lightweight and to avoid the additional data retention, security, and privacy considerations that come with persisting chat logs, especially since the Analysis Assistant handles data tied to a user's personal video uploads.

Q34. Could the chatbot leak one user's data to another user? *[Security]*

A: No. The Analysis Assistant's backend endpoint is protected by JWT authentication, and the database query that retrieves video/frame data explicitly filters by the authenticated user's ID (`WHERE user_id = :user_id`), not just by filename. This means even if two users uploaded videos with identical filenames, a user could never retrieve another user's analysis data, since the `user_id` filter is enforced at the query level, not just inferred from the conversation.

Q35. Could the AI 'hallucinate' and give a wrong explanation about a video's results? *[AI Safety / Limitations]*

A: This risk is mitigated by the RAG approach and the system prompt design — the assistant is explicitly instructed to use only the retrieved structured data (frame statistics, confidence scores, flagged regions) and never to invent data that wasn't provided to it. Since the underlying facts come from our own database rather than the LLM's general knowledge, the assistant's role is limited to phrasing and explaining real numbers in natural language, not generating new factual claims, which significantly reduces (though does not entirely eliminate) hallucination risk.

Section 7: Subscription & Payment Module

Q36. Why did you add a payment/subscription system to an academic FYP? *[Justification]*

A: The subscription model was added to address a real, practical concern: hosting AI inference and an LLM-based chatbot has ongoing infrastructure costs, even with free tiers. A tiered model (Free, Pro, Enterprise) allows the core detection functionality to remain free and accessible — which was always our primary objective — while offering an optional path for users who need higher usage limits to help sustain the platform's hosting costs beyond the academic project timeline. It also demonstrates a complete, production-style system rather than a purely academic prototype.

Q37. Why did you use Stripe instead of building your own payment processing? *[Technology Choice]*

A: Building custom payment processing would require PCI-DSS compliance, secure handling of raw card data, and significant security engineering effort that is both unnecessary and risky for a project of this scope. Stripe is an industry-standard, PCI-compliant payment processor — by using Stripe's hosted Checkout and billing portal, we never directly handle sensitive card information, which transfers most of the compliance and security burden to Stripe while still giving us full subscription management capability through their API.

Q38. Explain how a user's subscription status is kept in sync with Stripe. *[Technical Implementation]*

A: When a user completes checkout, Stripe redirects back to our application, and our confirm-session endpoint calls Stripe's API to verify the session and retrieve the resulting subscription details — plan, status, customer ID, and subscription ID — which are then persisted to the user's record in our users table. For ongoing changes (such as cancellations made directly within Stripe's billing portal), the user's dashboard reflects their current state by querying Stripe via the stored stripe_customer_id when needed, ensuring our database doesn't drift out of sync with the actual billing state held by Stripe.

Q39. What's the difference between your three subscription tiers, and how did you decide on that structure? *[Business Model — ACTION REQUIRED: confirm tier details]*

A: [Note to team: insert your actual finalized tier differences here — e.g., Free might include a limited number of analyses per month, Pro might include higher limits and priority processing, Enterprise might include API access or bulk processing. Be ready to explain the reasoning: e.g., 'Free tier ensures the core social-good objective of accessible detection remains available to everyone, while Pro/Enterprise target users with higher volume needs such as newsrooms or moderation teams.']

Q40. Does the subscription model conflict with your objective of free, open-source software? *[Justification]*

A: Not in our view — there's an important distinction between the codebase being open-source and the hosted service being free. The source code remains open and available for anyone to self-host or build upon at no cost, fulfilling our open-source and educational objectives. The subscription tiers apply specifically to our hosted, managed version of the platform, where ongoing infrastructure and AI inference costs exist regardless of licensing. This mirrors how many legitimate open-source projects operate a paid hosted offering alongside a free, self-hostable codebase.

Section 8: Use Cases, Testing & Quality Assurance

Q41. How did you ensure every use case is actually tested? *[Test Coverage]*

A: We maintained a direct traceability mapping between use cases and test cases throughout development. Every one of our 13 use cases (covering registration, login, video upload, analysis, viewing results, the four admin sub-functions, the two AI assistants, and the two subscription functions) has at least one corresponding test case in Chapter 7, and several — like registration — have both positive and negative test cases. This ensures no implemented functionality was left unverified.

Q42. Why did you use black box testing instead of white box testing? *[Testing Methodology]*

A: Black box testing evaluates the system from the user's perspective — inputs and expected outputs — without requiring knowledge of internal code structure. This approach was well-suited to our testing goals because it validates the system the way an actual user or examiner would interact with it, and it avoids tester bias that can occur when someone familiar with the implementation unconsciously tests around known weak points. We did, of course, also perform informal unit-level checks during development, but our formally documented test cases follow the black box approach.

Q43. What would happen if a user uploads a very large or corrupted video file? *[Edge Case Handling]*

A: The upload use case includes validation for file format (MP4, AVI, MOV) and file size (50MB maximum) before the file is accepted for processing. If a file exceeds the size limit or is in an unsupported format, the system rejects it at the validation stage with an appropriate error message, before any storage or processing resources are consumed. For corrupted files that pass initial format checks but fail during frame extraction, the processing pipeline is designed to catch such failures and report a processing error rather than crash or hang indefinitely.

Q44. How many test cases do you have, and were they all passing? *[Test Results]*

A: We have 17 documented test cases (TC01 through TC17) covering authentication (sign-up, negative sign-up, email verification, login, Google sign-in), the core detection flow (video upload, deepfake detection, viewing results), admin functionality (dashboard, analytics, user management, video management, activity log), the two AI assistants, and the two subscription functions. All 17 test cases are documented as passing in Chapter 7, reflecting the final, integrated state of the system at the time of submission.

Section 9: Admin Panel & System Management

Q45. What can an administrator do that a regular user cannot? *[Admin Features]*

A: The admin role has access to four additional capabilities not available to regular users:

- View System Analytics — total users, total videos, total analyses, deepfakes detected, pending analyses, and storage used.
- Manage User Accounts — view, search, and manage all registered users including their role and join date.
- Manage Videos — view all uploaded videos platform-wide and their analysis results, not just their own.
- View Activity Log — monitor recent platform activity including logins, uploads, and analyses across all users.

Q46. How do you prevent a regular user from accessing admin endpoints? *[Security]*

A: Every admin-restricted backend endpoint checks the authenticated user's role (extracted from their JWT token) before processing the request. If the role is not 'admin', the request is rejected regardless of whether the user is otherwise authenticated. This is enforced server-side on every request, not just hidden in the frontend UI, so a technically savvy user cannot bypass the restriction by directly calling the API.

Section 10: Limitations, Future Work & Project Reflection

Committees often end with these questions to assess self-awareness and critical thinking. Confident, honest answers here tend to leave a strong final impression.

Q47. If you had more time, what would you improve first? *[Future Work]*

A: [Note to team: personalize this with your genuine view, but a reasonable answer grounded in your Future Work section would be: 'We would prioritize audio deepfake detection, since voice cloning is an increasingly common attack vector that our current visual-only model cannot address, followed by exploring transformer-based architectures with GPU support for users who have access to that hardware and want higher accuracy.']

Q48. What was the most difficult technical challenge you faced? *[Reflection — ACTION REQUIRED: personalize]*

A: [Note to team: answer honestly based on your actual experience — common candidates based on the project's complexity include: integrating the AI model's frame-by-frame analysis with the asynchronous FastAPI request lifecycle without blocking the API; balancing detection accuracy against CPU-only inference speed; or coordinating the RAG pipeline's data retrieval and prompt construction for the Analysis Assistant to avoid hallucinated explanations.]

Q49. How does this project demonstrate the skills expected of a Software Engineering graduate? *[Outcome]*

A: The project required integrating skills across the full software engineering lifecycle: requirements analysis and UML modeling (use case, activity, sequence, class, and ER diagrams), full-stack web development (FastAPI backend, Next.js frontend, PostgreSQL database), applied machine learning (model selection, training considerations, and explainable AI techniques), systematic testing (black box test case design and traceability to use cases), and production-oriented concerns like authentication security, third-party payment integration, and scalable architecture design — reflecting both the academic rigor and practical engineering judgment expected at this level.

Q50. What would you do differently if you started this project again? *[Reflection — ACTION REQUIRED: personalize]*

A: [Note to team: this is a self-reflection question with no single correct answer — be honest. A safe, defensible answer structure: 'We would likely invest more time upfront in defining the database schema completely before implementation, since we had to extend the users and videos tables multiple times as new features (subscriptions, frame-level analysis) were added later — a more complete initial data model would have reduced rework.']

Section 11: Related Work & References — Quick Justification Sheet

Use this as a quick-reference cheat sheet if asked to justify why you cite or compare against specific prior work.

FaceForensics++ (ICCV 2019)

Purpose in our project: cited as a benchmark academic dataset/research effort for deepfake detection, used to contextualize the kind of manipulation types (face-swap, expression manipulation) our model needs to generalize across. Why it matters: it is one of the most widely cited datasets in deepfake detection literature, making it a credible reference point for comparing detection approaches.

Contextual Cross-Modal Attention (arXiv 2024)

Purpose in our project: cited in the Literature Review as an example of more recent multimodal detection research, used to justify why multimodal/cross-feature approaches are an active research direction, supporting our own discussion of multimodal analysis. Why it matters: shows awareness of recent academic trends beyond older foundational work, demonstrating the literature review goes beyond just citing the original 2019-era papers.

Multimodal Media Integrity Verification (IEEE Access 2023)

Purpose in our project: supports the academic grounding for combining multiple signal types (visual, possibly temporal) when verifying media authenticity, reinforcing why our system's architecture leaves room for future multimodal extension even though the current implementation is primarily visual/frame-based.

Microsoft Video Authenticator

Purpose in our project: used as a commercial/industry comparison point in our Comparative Analysis (Chapter 1 and Chapter 10), illustrating a tool built by a major technology company that nonetheless had restricted availability and limited transparency about its decision-making, supporting our motivation for building an accessible, explainable alternative.

Deepware Scanner

Purpose in our project: used as a second comparison point representing commercial/consumer-facing deepfake detection tools. Highlighted in our comparison table as lacking explainable AI outputs and operating as a more closed, less transparent system compared to our approach.

Section 12: Rapid-Fire & Curveball Questions

Short, punchy questions panels sometimes throw in to test composure and quick recall rather than depth. Keep these answers brief in the actual defense — 1-2 sentences, not paragraphs.

Q51. In one sentence, what does your system do?

A: It lets any user upload a video and receive an AI-generated, explainable verdict — with a confidence score, visual heatmap, and plain-language reasoning — on whether the video has been digitally manipulated.

Q52. What's your tech stack in 10 seconds?

A: Next.js frontend, FastAPI backend, PostgreSQL database, ConvNeXt-Tiny with PyTorch for 6-class detection, Groq for the AI chat assistants, and Stripe for subscriptions.

Q53. Is your system 100% accurate?

A: No system is 100% accurate, including ours — we report a confidence score rather than a binary guarantee specifically because detection is probabilistic, and we are transparent about that uncertainty rather than overstating our reliability.

Q54. Could someone bypass your detection by using a deepfake technique your model has never seen?

A: Potentially, yes — this is a known limitation of any supervised detection model, since it can only generalize to manipulation patterns related to what it was trained on. This is precisely why continuous retraining and dataset expansion is listed as part of our future maintenance considerations.

Q55. Why should we, as a committee, consider this project successful?

A: Because it doesn't just implement a deepfake classifier — it delivers a complete, accessible, explainable, and extensible system: from AI detection through to a usable interface, conversational explainability, administrative oversight, and a sustainable hosting model, all grounded in deliberate engineering decisions that we can justify and defend, as demonstrated throughout this document.

Final Notes Before Your Defense

- Fill in the three 'ACTION REQUIRED' answers (Q15 model accuracy, Q39 tier pricing, Q48/Q50 personal reflection) with your real, specific data before the defense — generic or placeholder answers here will be noticed.
- Practice saying these answers out loud in your own words rather than reading them verbatim — panels can tell the difference, and follow-up questions are easier to handle if you genuinely understand the reasoning rather than memorizing text.
- If you don't know the answer to something, it is far better to say "that's a good question, we did not specifically test/measure that, but our reasoning would be..." than to guess confidently and be wrong.
- Assign team members to different sections based on who worked on that part of the system, so each person can speak with genuine first-hand confidence about their area.